

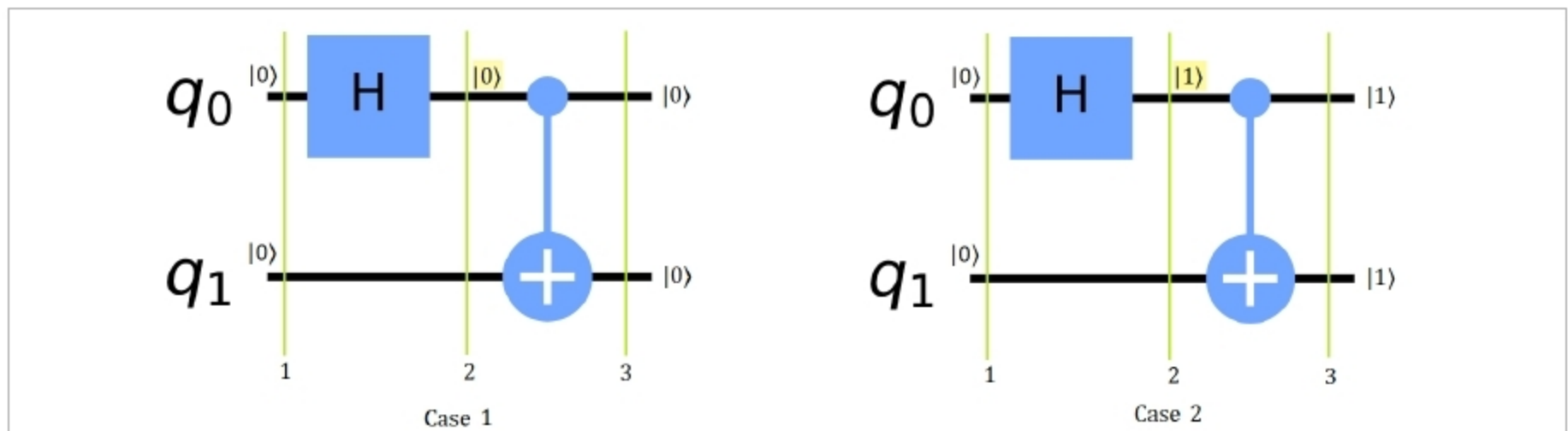


Figure 11: Stepwise display of the cases possible in an entanglement process

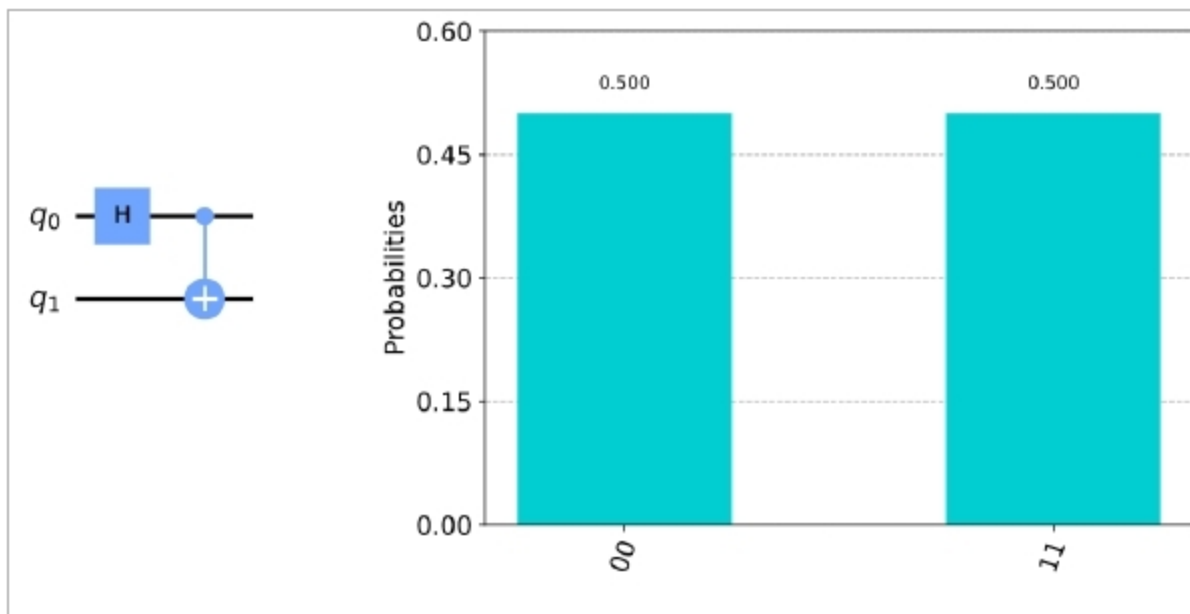


Figure 12: The entanglement circuit and the histogram showing equal probabilities of resultant states to be $|00\rangle$ or $|11\rangle$

$|1\rangle$, we can observe the two cases as displayed in Figure 11.

Figure 11 shows two cases of the circuit we have just built. We have marked stage 1-2-3 in both the figures for easy reference. The initial state (stage 1) of any qubit is $|0\rangle$ by default. Then we apply an H gate at q_0 . At stage 2, the state of q_1 is $|0\rangle$ but that of q_0 can be any of the two, i.e., $|0\rangle$ or $|1\rangle$.

In case 1, it is explained that if at stage 2, q_0 is $|0\rangle$ then the following CNOT gate will leave q_1 as it is and the final outcome (stage 3) will be $q_0 = |0\rangle$ and $q_1 = |0\rangle$, i.e., combined state $|00\rangle$.

Whereas, in case 2, it is explained that if at stage 2, q_0 is $|1\rangle$, then CNOT gate (which is controlled on q_0) will flip q_1 's state from $|0\rangle$ to $|1\rangle$. The final states of both qubits at stage 3 can be observed as $q_0 = |1\rangle$ and $q_1 = |1\rangle$, i.e., combined state of both qubits = $|11\rangle$.

This phenomenon, where we can see the effect of one qubit's state over the result of another, and the change in state of one due to change in state of the other, displays 'Entanglement'. The explanation proves that there's a 50-50 chance the result can be either $|00\rangle$ or $|11\rangle$. Let's

check that with the help of code:

```
backend = Aer.get_backend('statevector_simulator')
results = execute(qc, backend).result().get_counts()
plot_histogram(results, color='darkturquoise')
```

Figure 12 shows the entanglement circuit as well as the probabilities received after executing the circuit.

This marks the conclusion of this beginner-friendly practical tutorial of quantum computing using Python's Qiskit library. We hope the reader learnt some important concepts and will be intrigued to learn more. We encourage readers to learn by experimenting further, and have provided some great resources in the references section for those who are curious and interested. **END** 🐧

References

- [1] The Qiskit Textbook: <https://qiskit.org/textbook/preface.html>
- [2] Guidance from Qiskit India Challenge and IBM Quantum Challenge
- [3] Qiskit Global Summer School: <https://qiskit.org/learn/intro-qc-qh/>
- [4] Qiskit Documentation: https://qiskit.org/documentation/tutorials/circuits/3_summary_of_quantum_operations.html Qiskit
- [5] Circuit library: https://qiskit.org/documentation/apidoc/circuit_library.html

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